

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459914.77541 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1068 +/- 0.007
Transit depth (Rp/Rs)^2	1.14 +/- 0.15 [%]
Orbital Inclination (inc)	81.76 +/- 1.55
Airmass coefficient 1 (a1)	1.1356 +/- 0.0078
Airmass coefficient 2 (a2)	-0.1199 +/- 0.0022
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	None
Optimal Aperture	4.61
Optimal Annulus	10.22
Transit Duration (day)	0.1193 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.1068 ± 0.007 vs 0.1178 (deviation 1.31σ)
Duration fit vs published	0.1193 d vs 0.125 d (deviation 1.16σ)
Mid-time O–C	-3.0 min
Depth SNR	12.7
Residual scatter	0.68 %

Light curve

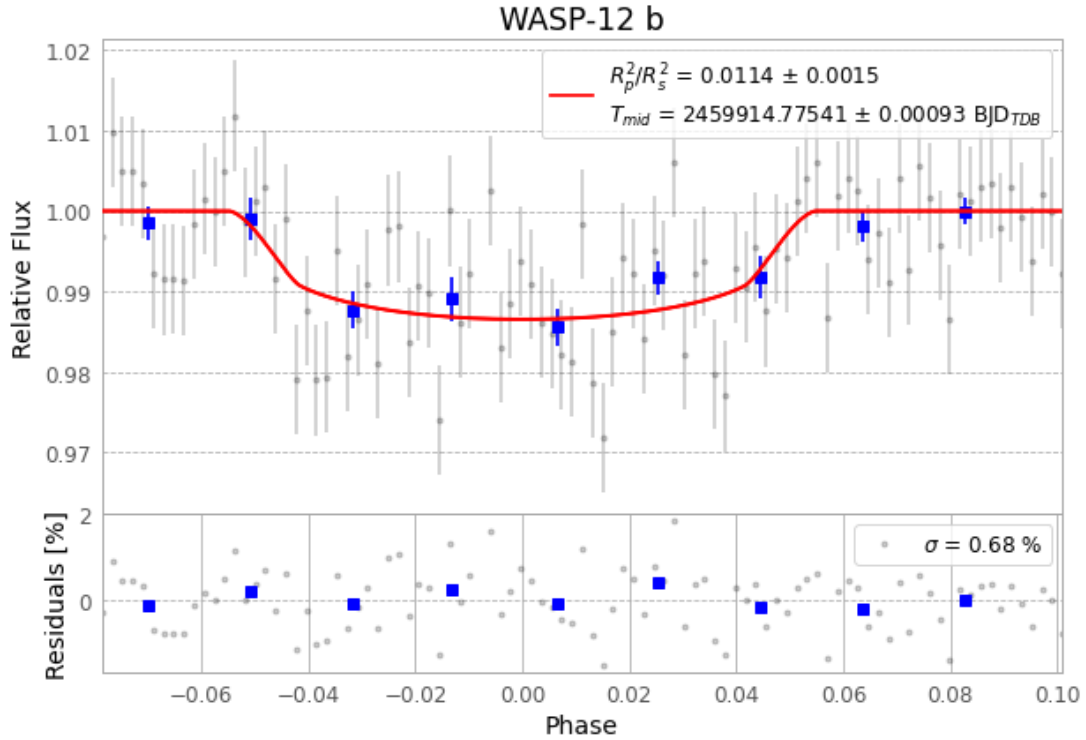


Figure 1 — FinalLightCurve_WASP-12 b_2022-11-30.png (SHA-256 10cc7fefb7a6a857...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 23], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd3b5648e62d0440...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

95 FITS frames (62 MB), WASP-12221201042415.FITS → WASP-12221201090615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221201.log (f61b70ac5e6a...), FinalLightCurve_WASP-12 b_2022-11-30.png (10cc7fefb7a6...), FinalLightCurve_WASP-12 b_2022-11-30.csv (b9fb250ab356...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 15:23Z.